The Air We Breathe

Pupil sheet

Name: _____ **Date:** _____

Safety: Breathe only into your own bottle, not onto the box; the teacher opens the window.

1. What I see on the screen

On the screen you see the **CO_2** card (carbon dioxide, in ppm — "parts per million") and its chart underneath. You start with **Start**; you move on with the green button at the bottom right, and the last button, a blue one, says **Finish**. You also type your prediction into the app before measuring. At each measurement you press **Capture**.

The sensor is slow in a particular way: it gives a new value once every **five seconds**, so the line climbs in steps. It is not broken.

The most important rule: when we measure the room's air, we **stay a step back** and do not lean over the box. Otherwise we measure our own air, not the room's.

2. What I do on my own

First I predict. I do not change my prediction after I measure.

1. If I breathe at the sensor, the number will: ☐ ↑ ☐ ↓ ☐ →
2. Once I stop, it will come back in: ☐ a second ☐ about ten seconds ☐ one to three minutes

because _____

_____

3. Put an arrow on the line where you think our room is today:

420	800	1200	2000 ppm
outdoors Excellent	Good	Moderate	Poor

While I wait for the recovery, I draw the shape of the line. How it went up, how it comes down.

Now I measure. If the number is too big for me to write, I just circle the word in the last column.

Moment	CO_2 (ppm)	Circle one
The room's air, at the start		Excellent / Good / Moderate / Poor
Just after I breathed		Excellent / Good / Moderate / Poor
After it came back		Excellent / Good / Moderate / Poor
After we opened the window		Excellent / Good / Moderate / Poor
A second breath, the same		close to the first? yes / no

Working it out. The biggest: _____ ppm the smallest: _____ ppm
How long the recovery took: _____ minutes and _____ seconds. Children in the room: _____

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

It went up fast and came down slowly because _____

_____

Our class rule: when, and who, opens the window? Use your numbers.

_____

Exit ticket

Write one thing that surprised you today.

_____